

Nebulised therapies for smoke inhalation injury: mechanistic rationale, evidence gaps, and the case for an *adaptive platform trial*.

Dr Ganesh Hanumanthu, FRCA FFICM FRCP RCPaME MAcadMedEd
Consultant Anaesthetist & Intensivist · Mersey & West Lancashire Teaching Hospitals NHS Trust (Whiston Hospital)
Senior Clinical Lecturer & Co-Director MBChB Years 3 & 4 · Lancaster Medical School, Lancaster University

BBA 2026 · BOARD P111 · 12:51–12:54
CORRESPONDENCE
Ganesh.Hanumanthu@merseywestlancs.nhs.uk

01 · SCALE OF THE PROBLEM

Smoke inhalation injury is the *single largest preventable driver* of burn mortality.

35%
BURN ADMISSIONS WITH INHALATION INJURY¹

38%
SUBSEQUENT PNEUMONIA RATE¹

+20%
ABSOLUTE MORTALITY INCREASE¹

ZERO
THERAPIES OF PROVEN BENEFIT²

02 · THE GAP

Treated almost entirely by guesswork.

Despite widespread use of nebulised unfractionated heparin, N-acetylcysteine, and β -agonists in specialist burn centres, the underpinning evidence remains **fragmented, underpowered, and inconclusive**.²

Inhalation injury sits at the intersection of small disease incidence, heterogeneous severity, and high acuity — conditions in which traditional two-arm RCTs have systematically failed.

03 · TWO-HIT PATHOPHYSIOLOGY

A distinct *two-hit* injury cascade — a mechanistic rationale for each therapy.

FIRST HIT · 0–6 HOURS
Chemical ciliotoxicity

INTACT

DENUDED

- Direct chemical injury to epithelium
- Ciliary destruction + denudation
- Loss of mucociliary clearance
- Neutrophil influx + cytokine storm

→ Ref 2

SECOND HIT · 12–72 HOURS
Fibrin-cast obstruction

FIBRIN-RICH AIRWAY CAST

- Bronchial blood flow $\uparrow$ 10-fold
- Plasma exudation → coagulopathy
- Fibrin + cellular debris casts
- Small-airway obstruction · V/Q mismatch

→ Ref 2

NEBULISED THERAPEUTIC TARGETS

Three mechanistic anchors

Nebulised unfractionated heparin
Attenuates second-hit fibrin coagulopathy^{2,3}

N-Acetylcysteine
Mucolytic + antioxidant; pairs with heparin³

β -agonist (albuterol)
Bronchodilation + mucociliary clearance

04 · AIM

Why this synthesis exists.

To synthesise contemporary preclinical, translational, and clinical evidence for nebulised **heparin, NAC, and β -agonists** in smoke inhalation injury.

01 Mechanistic plausibility — does the biology hold up?

02 Methodological limits — why have RCTs failed?

03 Trial design — what methodology could finally answer?

05 · METHOD

Narrative synthesis approach.

Search. PubMed, Embase, Cochrane (inception–April 2026); inhalation/smoke injury × each candidate therapy.

Inclusion. Observational cohorts, RCTs, and existing SR/meta-analyses in adult burn populations.

Appraisal. Studies appraised narratively — no quantitative pooling (heterogeneity precludes it).

Synthesis. Findings grouped by intervention and trial-design limitation.

06 · WHY RCTS FAIL

Four structural barriers — together, fatal.

LOW INCIDENCE
~5,000 UK cases/yr
~25 specialist centres

HETEROGENEITY
Severity, TBSA, timing, co-morbidity

CONFOUNDED CO-INTERVENTION
Heparin always given with

HIGH ACUITY

FAILURE to recruit

HEPBURN⁴: powered for 116, closed at **n = 13**.

07 · THE CLINICAL EVIDENCE BASE

A 27-year evidence base — *and a flagship RCT that collapsed under its own weight*.

STUDY (YEAR)	FINDING	STATUS
Desai 1998	Heparin + NAC $\downarrow$ atelectasis, reintubation, mortality (paediatric retrospective).	↓ MORTALITY
Enkhbaatar 2007	Aerosolised anticoagulants $\downarrow$ acute lung injury (ovine burn+smoke).	PRECLIN. SIGNAL
Miller 2009	Heparin/NAC $\uparrow$ LIS, oxygenation, survival (n=30 adults, retrospective).	↓ MORTALITY
Elsharnouby 2014	10,000 IU vs 5,000 IU heparin: higher dose $\downarrow$ LIS & vent duration (n=29).	DOSE SIGNAL
Glas 2014–2020 · HEPBURN	International RCT. Powered 116. Stopped at n=13 .	TERMINATED

STUDY (YEAR)	FINDING	STATUS
McIntire 2017	Larger HIHI cohort; signal blunted vs Desai/Miller.	EQUIVOCAL
Glas 2016 IPD-MA	All ventilated ICU; no convincing benefit on propensity-matched.	NULL ON IPD
Lan 2020 MA	9 studies, none multicentre RCT. Mortality $\downarrow$ (RR 0.75); VFD $\uparrow$.	MODEST ↓ MORT
Palmieri 2006	Continuous nebulised albuterol $\uparrow$ oxygenation, $\downarrow$ lung water (ovine).	PRECLIN. SIGNAL
Hakim 2025	n=88. $\uparrow$ VFDs, $\uparrow$ PaO ₂ /FiO ₂ ; no mortality difference.	↑ VFDS · NO Δ MORT

Twenty-seven years of investigation. Preclinical signal: strong. Clinical evidence: heterogeneous, single-centre, retrospective, and — when scaled up — *unable to recruit*.

08 · THE ADAPTIVE PLATFORM SOLUTION

A single, perpetual platform — answering many questions at once.

DOMAIN 1
Heparin

DOMAIN 2
NAC

DOMAIN 3
 β -agonist

DOMAIN 4+
future agents

PLATFORM SPINE · CONTINUOUS ENROLMENT

BAYESIAN ADAPTATION

Drop futile arms

Up-randomise to winning

Add new agents

Subgroup learning

Master protocol, multi-centre. Patients consented once into the platform; randomised across domains. Bayesian updating drops futile arms early. REMAP-CAP proof of concept: 13,000+ patients enrolled globally during COVID-19.⁶

09 · CONCLUSION

From mechanism to method.

"Traditional RCTs have not resolved — and arguably *cannot resolve* — uncertainty in inhalation injury. Bayesian adaptive platform trials offer a pragmatic pathway to definitive, practice-changing evidence."

FOR THE CLINICIAN

Heparin, NAC, and β -agonists remain reasonable adjuncts in specialist centres on mechanistic grounds — but practice should be acknowledged as evidence-light. Local protocols should support future platform-trial participation.

FOR THE BBA COMMUNITY

The UK burn-care network — concentrated, well-coordinated, and clinically connected — is uniquely placed to lead an international adaptive platform trial. This is the design that can finally close a 27-year evidence gap.

PROPOSAL

A UK-led *adaptive platform trial* for nebulised therapies in smoke inhalation injury — embedded across the BBA-affiliated specialist burn centre network.

PLATFORM DESIGN

Master protocol; three initial domains (heparin, NAC, β -agonist); Bayesian adaptive randomisation; pre-specified futility / efficacy stopping rules.⁶

PRIMARY OUTCOME

28-day ventilator-free days (compatible with HEPBURN⁴ & Hakim 2025⁵); pneumonia, mortality, ICU LoS as secondary.

FEASIBILITY ADVANTAGE

One consent, one infrastructure, one statistical core. Smaller per-domain samples; perpetual recruitment overcomes the HEPBURN failure mode.

FULL BIBLIOGRAPHY & DOIS

Scan for the complete reference list, with direct links to PubMed and DOI.

All 20 references verified · maintained alongside the live poster microsite



SCAN WITH PHONE CAMERA